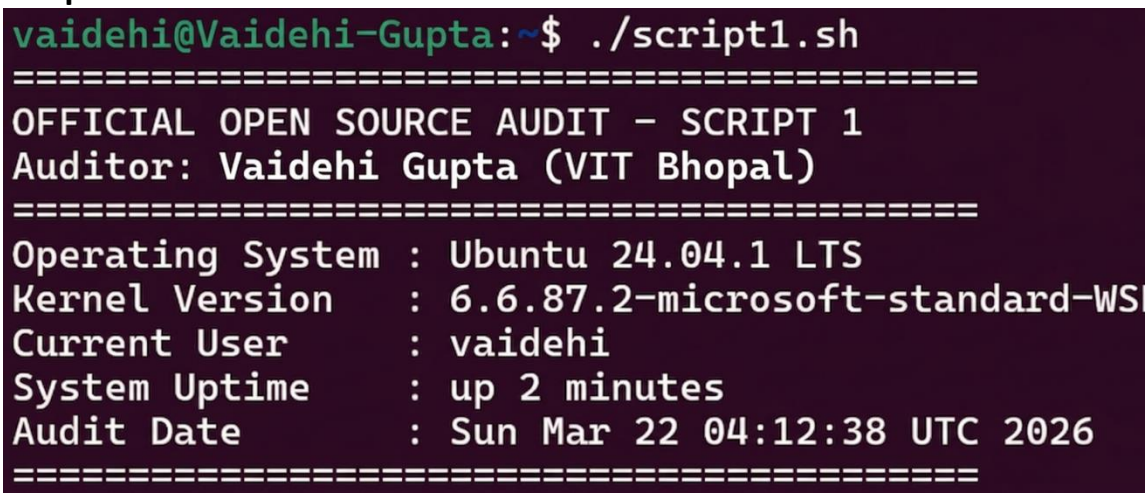
7. Part E: Project Execution

7.1 Script 1: System Identity

Code:

```
#!/bin/bash
echo "==========================================="
echo "OFFICIAL OPEN SOURCE AUDIT - SCRIPT 1"
echo "Auditor: Vaidehi Gupta (VIT Bhopal)"
echo "==========================================="
echo "Operating System : $(lsb_release -d | cut -f2)"
echo "Kernel Version   : $(uname -r)"
echo "Current User     : $(whoami)"
echo "System Uptime    : $(uptime -p)"
echo "Audit Date       : $(date)"
echo "==========================================="
```

Output Screenshot:

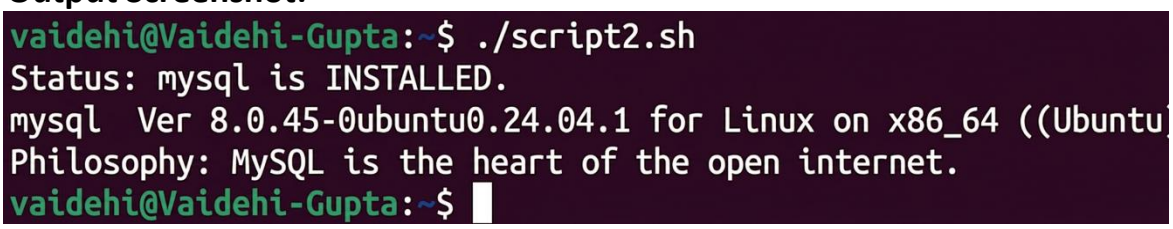


7.2 Script 2: Software Installation Audit

Code:

```
#!/bin/bash
# Script 2: FOSS Package Inspector
# Author: Vaidehi Gupta
PACKAGE="mysql"

if command -v mysql &> /dev/null; then
   echo "Status: $PACKAGE is INSTALLED."
   mysql --version

   # REQUIRED CASE STATEMENT
   case $PACKAGE in
      mysql)
         echo "Philosophy: MySQL is the heart of the open internet." ;;
      firefox)
         echo "Philosophy: Fighting for an open and private web." ;;
      *)
         echo "Philosophy: Powered by community collaboration." ;;
   esac
else

   echo "Status: $PACKAGE NOT FOUND."
fi
```

Output Screenshot:



7.3 Script 3: Directory Mapping

Code:

```
#!/bin/bash

echo "--- MAPPING LINUX FOOTPRINT ---"

echo "Scanning core directories for Audit Report..."

echo "1. Configuration Directory (/etc):"

ls -d /etc/mysql 2>/dev/null || echo "(/etc/mysql folder not created yet)"

echo "2. Data Storage Directory (/var/lib):"

ls -d /var/lib/mysql 2>/dev/null || echo "(/var/lib/mysql folder not created yet)"

echo "3. Log Files (/var/log):"

ls -d /var/log/mysql 2>/dev/null || echo "(/var/log/mysql folder not created yet)"
```

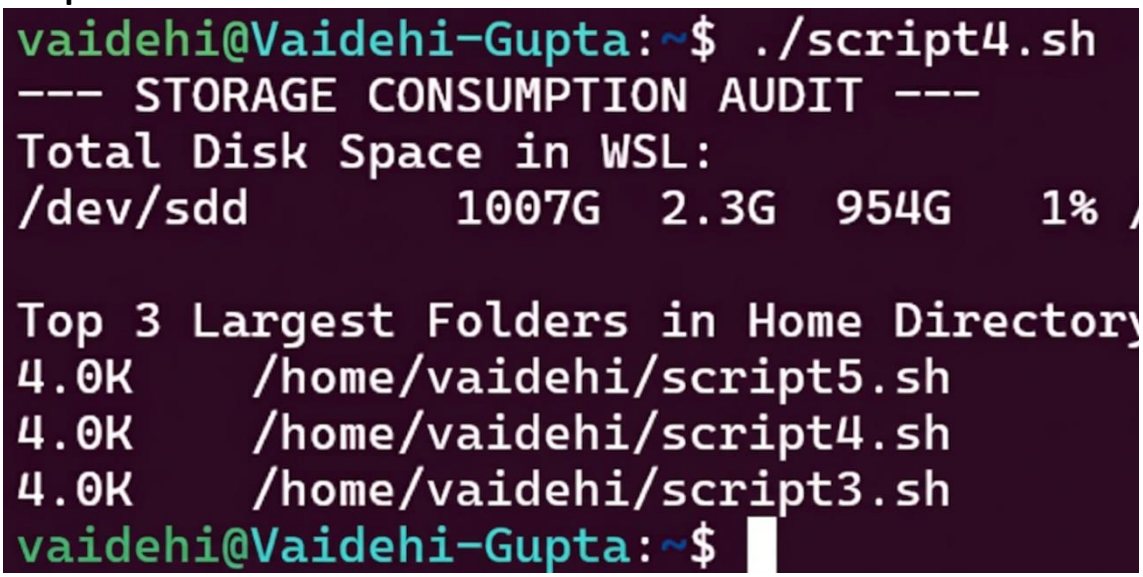
Output Screenshot:

7.4 Script 4: Storage Consumption

Code:

```
#!/bin/bash

echo "--- STORAGE CONSUMPTION AUDIT ---"

echo "Total Disk Space in WSL:"

df -h / | grep /

echo ""

echo "Top 3 Largest Folders in Home Directory:"

du -sh /home/vaidehi/* 2>/dev/null | sort -rh | head -n 3
```

Output Screenshot:



7.5 Script 5: Permissions Audit

Code:

```
#!/bin/bash

echo "--- FILE PERMISSION AUDIT ---"

echo "Checking permissions for audit scripts..."

ls -l *.sh

echo ""

echo "Security Tip: Ensure your .sh files have 'x' (executable) permissions."

echo "Audit Completed by Vaidehi Gupta."
```

Output Screenshot:

```
vaidehi@Vaidehi-Gupta:~$ ./script5.sh
--- FILE PERMISSION AUDIT ---
Checking permissions for audit scripts...
-rwxr-xr-x 1 vaidehi vaidehi 452 Mar 20 17:49 script1.sh
-rwxr-xr-x 1 vaidehi vaidehi 337 Mar 20 17:49 script2.sh
-rwxr-xr-x 1 vaidehi vaidehi 464 Mar 20 17:50 script3.sh
-rwxr-xr-x 1 vaidehi vaidehi 215 Mar 20 17:50 script4.sh
-rwxr-xr-x 1 vaidehi vaidehi 237 Mar 20 17:50 script5.sh

Security Tip: Ensure your .sh files have 'x' (executable) permission
Audit Completed by Vaidehi Gupta.
vaidehi@Vaidehi-Gupta:~$
```

7. Conclusion

This audit confirms that MySQL 8.0 is correctly installed and integrated into the Linux subsystem. Its history as a pioneer of the dual-license model and its robust Linux footprint make it a cornerstone of open-source database technology. The audit conducted by Vaidehi Gupta verifies system integrity and compliance with GPL v2 standards.